

CircuLeak Industrial Carbon Audit

Executive Decision-Support Report — Facility: **HM Industries**
Audit Date: September 13, 2026 | Sector: Automotive Component Casting

1. Executive Summary

An industrial carbon leak and circular diagnostic was executed for HM Industries . Analysis of 2352 verified telemetry records indicates plant emissions of 104,079 kgCO2e, operating at an intensity of 0.2 kgCO2e per unit of production (-86.7% vs sector benchmark). 4 behavioral anomaly leaks were identified, with primary dissipation observed in Aluminum Induction Melting Furnace. Deployment of 3 prioritized circular interventions requires an estimated investment of INR 1,725,000, delivering INR 750,000 in recurring annual savings with a payback period of 2.3 years and reducing plant footprint by 85,200 kgCO2e.

2. Facility Profile

Business Name	HM Industries	Sector	Automotive Component Casting
Location	Industrial Corridor, India	Production Type	General Manufacturing & Precision Forming
Production Volume	25,000 units	Operating Hours	16.0 hrs/day
Employees	120	Energy Sources	grid_electricity, natural_gas

3. Emission Overview

Metric	Value	Benchmark & Regulatory Context
Total Emissions	104,079.3 kgCO2e (104.08 tCO2e)	Total verified operational footprint
Emission Intensity	0.2 kgCO2e/unit	Sector Avg: 1.5 (Best in Class (Top 10%))
CCTS Regulatory Threshold	1.3 kgCO2e/unit	Status: Compliant
Total Electricity	127,273.5 kWh	Grid Factor: 0.716 kgCO2e/kWh (India CEA v19)

4. Top Carbon Leaks (Structural Hotspots)

Rank	Equipment	Process	Emissions (kgCO2e)	% Total
1	Aluminum Induction Melting Furnace	Melting & Holding	47,270.4	45.4%
2	Solution & Aging Heat Treatment Furnace	Heat Treatment & Quenching	21,091.1	20.3%
3	Screw Air Compressor (100 HP)	Compressed Air Utility	11,085.1	10.7%
4	Die Cooling Water Circulation System	Cooling Utility	8,665.4	8.3%
5	HPDC Die Casting Cell #01 (800T)	High Pressure Die Casting	7,138.6	6.9%

5. Behavioral Anomalies Detected

Equipment	Period	Risk	Baseline	Observed	Diagnosis
Aluminum Induction Melting Furnace	22:00-06:00 (Off-Hours / Non-Production)	86.2	0.00	191.89	Aluminum Induction Melting Furnace averages 191.9 kWh during non-production hours when baseline expectation is near zero.

Die Cooling Water Circulation System	22:00-06:00 (Off-Hours / Non-Production)	85.1	0.00	36.07	Die Cooling Water Circulation System averages 36.1 kWh during non-production hours when baseline expectation is near zero.
Aluminum Induction Melting Furnace	23:00-00:00	71.2	185.62	268.90	Energy spike of 268.9 kWh observed (44.9% above 185.6 kWh baseline).
Screw Air Compressor (100 HP)	22:00-06:00 (Off-Hours / Non-Production)	66.2	0.00	28.98	Screw Air Compressor (100 HP) averages 29.0 kWh during non-production hours when baseline expectation is near zero.

6. Intervention Priority & Financial ROI

#	Intervention	CO2 Saved (kg)	Investment (INR)	Annual Savings (INR)	Payback	Feasibility
1	Fuel Switching: Coal to Agro-Residue Biomass Briquettes	42,000	INR 480,000	INR 210,000	2.29 yrs	Medium
2	Industrial High-Temperature Heat Pump for Process Water	34,000	INR 1,200,000	INR 420,000	2.86 yrs	Medium
3	Ultrasonic Compressed Air Leak Audit & Sealing	9,200	INR 45,000	INR 120,000	0.38 yrs	High
4	Variable Frequency Drive (VFD) Retrofit on Air Compressor	14,600	INR 220,000	INR 135,000	1.63 yrs	High

7. Circularity Score & 5-Year Decarbonization Trajectory

Circularity Rating: Current Score = **69.2/100** (B). Projected Score post-interventions = **94.2/100**.

Year	Baseline (kgCO2e)	With Actions (kgCO2e)	Annual Reduction	Cumulative Savings
2026	104,079	69,149	34,930 kg	INR 309,750
2027	105,640	36,974	68,666 kg	INR 929,250
2028	107,225	37,529	69,696 kg	INR 1,814,250
2029	108,833	38,092	70,742 kg	INR 2,743,500
2030	110,466	38,663	71,803 kg	INR 3,717,000